



The effect of wake-closure phenomenon on gas atomization performance

Jason Ting*, Michael W. Peretti, William B. Eisen

Crucible Research, Pittsburgh, PA 15205-1022, USA

Abstract

The relationship between melt flow rates, gas-only aspiration of the atomizer nozzle and operating pressures are elusive, and most often these parameters are interdependent. The understanding of these interactions, however, is crucial to the efficient production of fine metal powder. Atomization yield above and below a condition known as wake closure pressure can produce median powder size (D_{50}) variation as large as 42%. Here, a pulsatile atomization model is presented to elucidate the interaction between the atomizing gas and the melt around the melt pour tube. The model reveals that melt flow rate is determined by the time-averaged stagnation pressure formed in front of the melt orifice. This stagnation front dictates the behavior of the melt exiting the melt orifice and how it interacts with the atomizing gas. It is noted, however, that the wake-closure phenomenon is highly sensitive to atomizer and melt pour tube geometry and it possesses a hysteresis type aspiration behavior that suggests that there is an energy of formation associated with the wake-closure structure. Furthermore, the atomizing gas has to possess the necessary energy to overcome this energy barrier in order to form the wake closure structure. Concurrently, the entropy of the system increases with the formation of Mach disk and oblique shocks, leading to a lower equilibrium state for the gas dynamics. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The gas atomization process in powder metallurgy has been described as the disintegration of a liquid mass into a collection of liquid metal droplets by the impact of high velocity gas flow. The liquid metal droplets subsequently spheroidize, cool and solidify into metal powder particles, which can typically range from 1 μm to 1 mm in diameter. The spherical metal alloy powders produced by the gas atomization process have very attractive material properties because of the inherent high cooling rate of this process. Furthermore, the spherical powder morphology is also very desirable for the growing metal injection molding industry. One of the main advantages of the gas atomization process is that it can produce many solidification benefits such as particle microstructure refinement, which is unattainable by conventional ingot metallurgy. This refinement is particularly beneficial to highly alloyed materials

such as tool steels and nickel-based superalloys, where the alloy microstructure has tremendous influence on the material properties.

In a typical gas atomization process, the molten metal stream flows from the bottom of a heated tundish through a melt feed tube into the atomization zone. Here, the liquid is disintegrated into a collection of liquid droplets as it encounters a high velocity gas stream just below the exit of the melt orifice. Depending on the particular design, there are generally two types of gas atomization configurations that are used in low and moderate pressure powder metallurgy production. These are known as: (1) open-stream or free-fall; and (2) closed, confined-feed, or close-coupled atomization [1]. For applications where fine powder yields are not necessary, a free-fall melt configuration is adequate. In this situation, the unconstrained stream of molten metal falls under the influence of gravity from the outlet of the tundish through the atomizing gas. Due to the absence of liquid melt stability and concentricity, the free-fall gas atomization configuration is unsuitable as an efficient gas atomization technique. On the other

* Corresponding author.

E-mail address: jason@cruciblresearch.com (J. Ting).

hand, the confined-feed gas atomization, where the gas jets impinge on the pour tube prior to merging at a focal point located in front of the orifice of the pour tube, provides for a more efficient configuration to transfer the kinetic energy of the atomizing gas to the melt stream for producing finer powders. Therefore, the desire to achieve a greater yield of ultra-fine powders (diameter $< 20 \mu\text{m}$) with improved efficiency at high to moderate gas pressure seems to promote the use of confined-feed atomization nozzle in powder metal production industries [1–7].

Even though the confined-feed nozzles are prevalent in powder metal production industries and are acknowledged as the most efficient gas atomization configuration for fine powder production, the operation of a confined-feed nozzle is still generally regarded as a difficult ‘art’. A desired operational characteristic of the confined-feed nozzle configuration is that subambient pressure near the melt orifice is needed for stable atomization. The rapidly flowing gas passing the periphery of the melt tip can produce an overambient or subambient pressure region in front of the delivery tube orifice that is often known as the aspiration base pressure. Melt freeze-off can arise in high pressure gas atomization, when the aspiration base pressure is near or above ambient pressure at the melt pour tube orifice. Under such high pressure conditions, the stability at the melt tip can better be controlled, if the atomizer nozzle and melt tip are designed to experience a subambient pressure. Operating in this intrinsic aspiration regime, the subambient pressure at the orifice should provide for a more stable melt feed by drawing the melt stream towards the atomization zone.

Consequently, the melt orifice aspiration base pressures have generally been used as an indicator for evaluating the atomization stability of a gravity-fed, confined-feed atomizer nozzle. The majority of the nozzles currently used in producing metal powders belong to the High Pressure Gas Atomizer (HPGA) type nozzles with an annular-slit or discrete jets. Numerous researchers have indicated that atomizing at the deep gas-aspiration pressures (at condition of wake-closure) in close coupled gas atomizers yields the finest atomized powder particle distribution [2–6]. On the other hand, Miller et al. concluded that the atomization performance is, at best, weakly related to aspiration [7]. But none has provided a satisfactory model that could explain these observations. The causes and influences of the wake-closure are investigated and presented here in an pulsatile atomization model that could readily explain many of the atomization observations described in this work and elsewhere. The relationship between measured deep aspiration generated by an industrial gas atomizer nozzle and melt flow rate with observed particle size distribution will be given.

2. Experimental

Seven 22.7 kg heats of a Ni-based alloy were atomized. Prior to atomizing each heat, the gas-only aspiration profile of the atomizer nozzle was measured with a digital pressure transducer to determine the manifold pressure at which wake-closure occurred. The digital data were acquired through a data logger and stored in a portable laptop PC. Two pour tube set lengths, which represent the length of the pour tube protrusion from the base of the atomizer nozzle, were investigated in this study. The aspiration base pressure for the shorter pour tube can be seen in Fig. 1, and that for the longer pour tube can be seen in Fig. 2. The aspiration measurements from the atomizer nozzle with the longer pour tube indicated that the wake-closure pressure (hereafter known as WCP) developed around 4.93–4.95 MPa (715–718 psi); this was highly repeatable. In this study, only the atomizer nozzle with the longer pour tube configuration will be used in atomization heats. For each atomization heat of the Ni-based alloys, the atomization gas pressure in the nozzle manifold was selectively tuned to atomize at pressures either above, or below the WCP. Heats that were atomized below WCP had shallow gas-only aspiration of 0.25 atm (3.7 psi) below atmosphere pressure, while heats atomized above WCP had deep gas-only aspiration of 0.58 atm (8.5 psi) below atmosphere. The atomization heats were timed. The atomization pressure was monitored, and the particle size distribution was vibratory screened and sonically sieved to obtain the particle size distribution above and below $45 \mu\text{m}$, respectively.

3. Results

The particle size distributions from this study are given in Fig. 3 and the atomization results are tabulated in Table 1. From the table, one found that the heats that are atomized at pressures above WCP produced powders that have dramatically smaller median particle size than those heats atomized at pressures below WCP (Fig. 3). In particular, the median particle size from L1270 that was atomized 0.017 MPa above WCP was 42% (or, 26.5 micron) smaller compared with that for L1271 that was atomized 0.034 MPa below WCP. This particle refinement in L1270 is a result of the heat having a 60% longer atomization time, translating to a 64% greater GMR, than L1271 (Fig. 4). In addition, atomizing at pressures just above WCP also was associated with a 23% improvement in particle size standard deviation (D_{84}/D_{50}) compared with that atomized below WCP (Fig. 5). Atomizing above WCP gave a narrower particle size distribution. Seeing that the gas atomization pressures for both cases differed by less than 1.5%, the observed improvement in particle refinement cannot

possibly be attributed to an increase in atomizing pressures and gas kinetics. Results from this study indicated that the observed metal flow rate was not behaving as what was earlier assumed, namely that, deeper aspiration enhanced metal flow rate. One can assert that the

observed change in melt atomization is likely due to a modification in the aspiration gas dynamics during atomization.

Unlike the wake closure pressure for the long pour tube length, the wake closure pressure for the geometri-

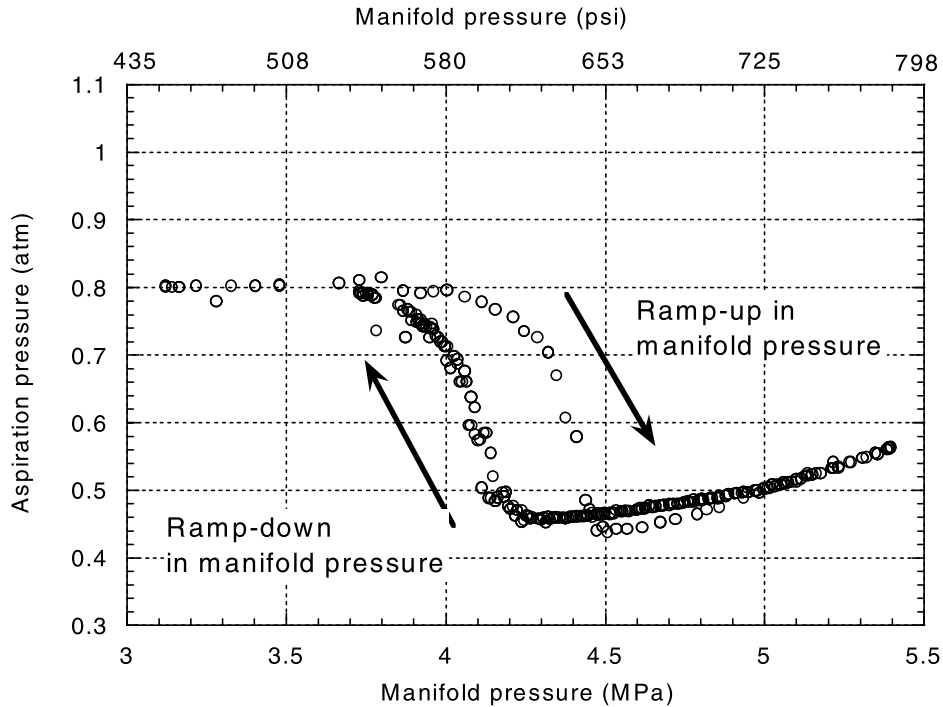


Fig. 1. Gas only aspiration measurement with short pour tube length showing wake-closure hysteresis.

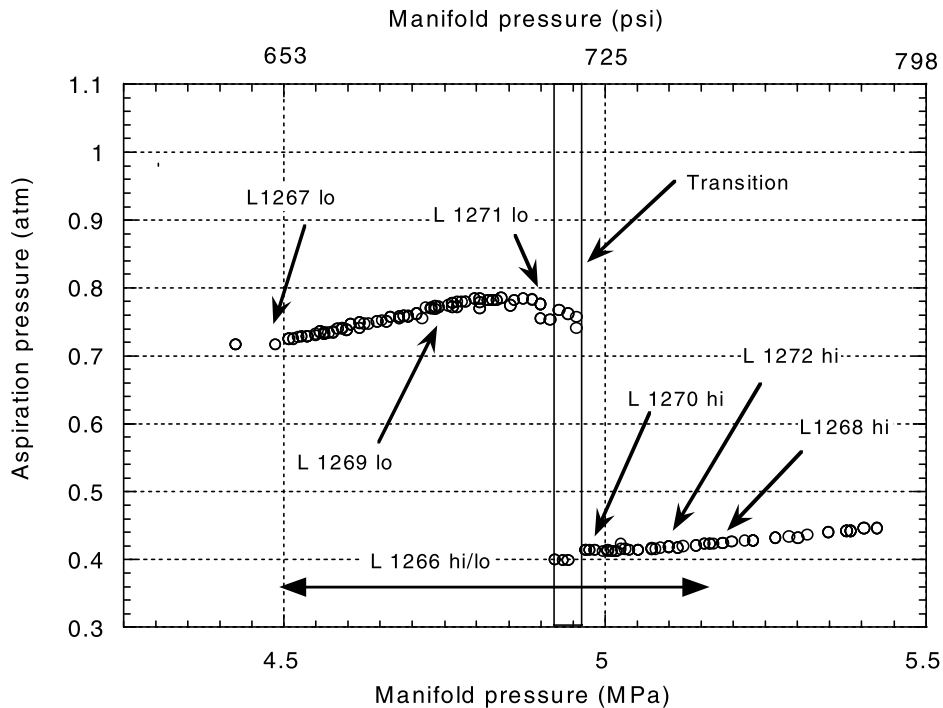


Fig. 2. Gas-only aspiration measurement with long pour tube length showing the location, where the wake-closure occurs. Also shown are the pressures at the Heats are atomized. (Reproduced with modification from [8]).

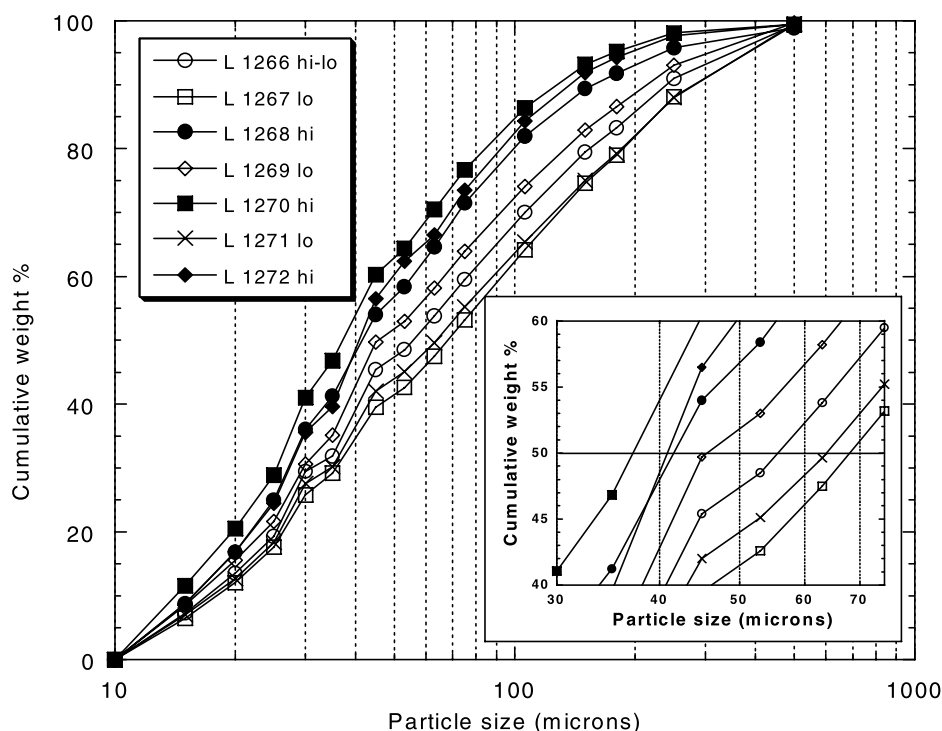


Fig. 3. Powder size distribution. (Reproduced with modification from [8]).

Table 1
Atomization parameters and results for a nickel-based alloy

Heat	GMR ^a	Atomizing pressure (MPa)	Run time (s)	AMFR ^b (kg min ⁻¹)	D_{50} (μm)	Std. deviation ^c D_{84}/D_{50}
L1266	2.46	4.48–5.17	60	22.7	55.4	3.30
L1267	1.77	4.48	73	27.7	68	3.16
L1269	2.35	4.69	61	22.3	45.1	3.48
L1271	2.48	4.89	59	23.2	63.5	3.38
L1270	4.06	4.96	94	14.5	37	2.61
L1272	4.04	5.10	88	15.5	40.5	2.57
L1268	3.65	5.17	83	16.4	41.5	2.82

^a GMR, gas-to-melt ratio (kg kg⁻¹).

^b AMFR, average melt flow rate.

^c Std. dev. (D_{84}/D_{50}), standard deviation.

cally shorter pour tube was more intriguing (Fig. 1). The wake closures for the short pour tube happened at a lower pressure than that for the long pour tube. The observed lower wake closure pressure with the shorter pour tube length was in fact of no surprise, because this was recently reported by Wolf et al. [9]. In this study, however, two wake closure pressures are observed in the aspiration measurement for the shorter pour tube configuration. The wake closure phenomena were very gradual with very minimal discontinuity compared with that observed for the long pour tube configuration (Fig. 2). During manifold pressure ramp up, the first wake closure pressure was observed at 4.35 MPa, and during manifold pressure ramp down, the second wake closure pressure was observed at 4.1 MPa.

In what follows, a gas dynamic model elucidating the cause of this observed particle refinement, that takes into account of the wake closure phenomenon, will be presented.

3.1. The wake-closure phenomenon

The HPGA type nozzle configuration contains the basic flow characteristics seen in air-breathing truncated plug nozzle studies by Sule and Mueller [10] for ‘altitude-compensating’ operation. In order to understand the wake-closure phenomenon, and to answer the cause of the apparent augmented particle refinement during melt atomization across the WCP, events of the gas dynamics in the recirculation zone and the surrounding gas field had to be modeled. (In the pa-

per, the term recirculation zone and wake will be used interchangeably).

The gas flow dynamics of the internal wake of an HPGA can be represented as a control volume previously described by Ting et al. [11] (Fig. 6). Although the proposed two-dimensional triangular control volume is an over simplification of a real three-dimensional recirculation zone, it adequately serves to model

the general events in the internal wake. According to the control volume approach, the gas enters the wake through the stagnation front and exits along the circumferential edge of the melt tip base. It has been disclosed [11] that the radial velocity component of the atomizing gas at the stagnation front is a function of the apex angle of the atomizer, and it strongly influences the stagnation pressure such that large radial

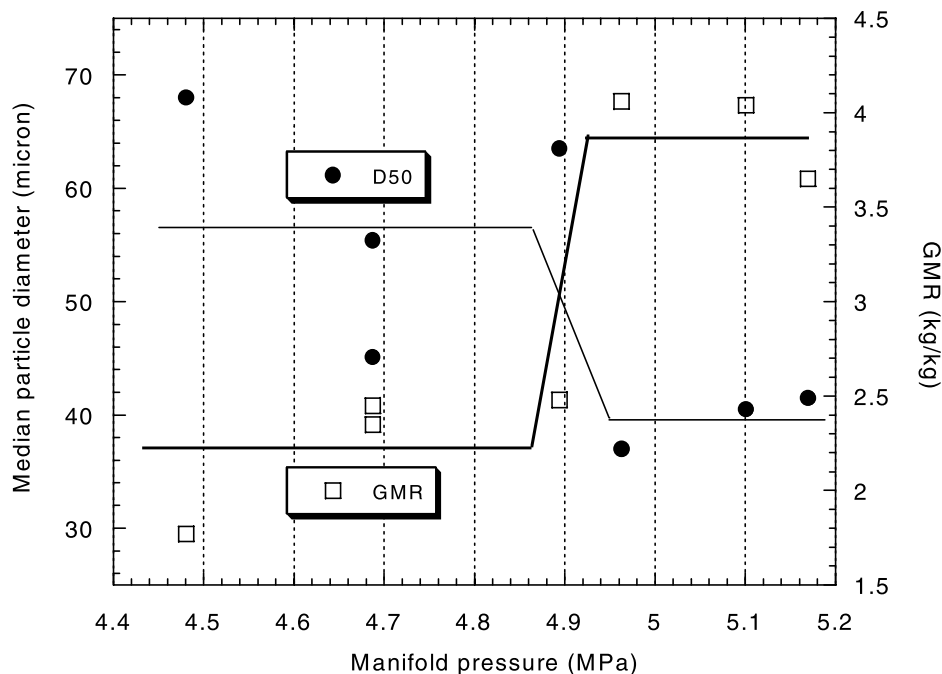


Fig. 4. Variation in median particle diameter and GMR versus manifold pressure measured across the WCP. (Reproduced with modification from [8]).

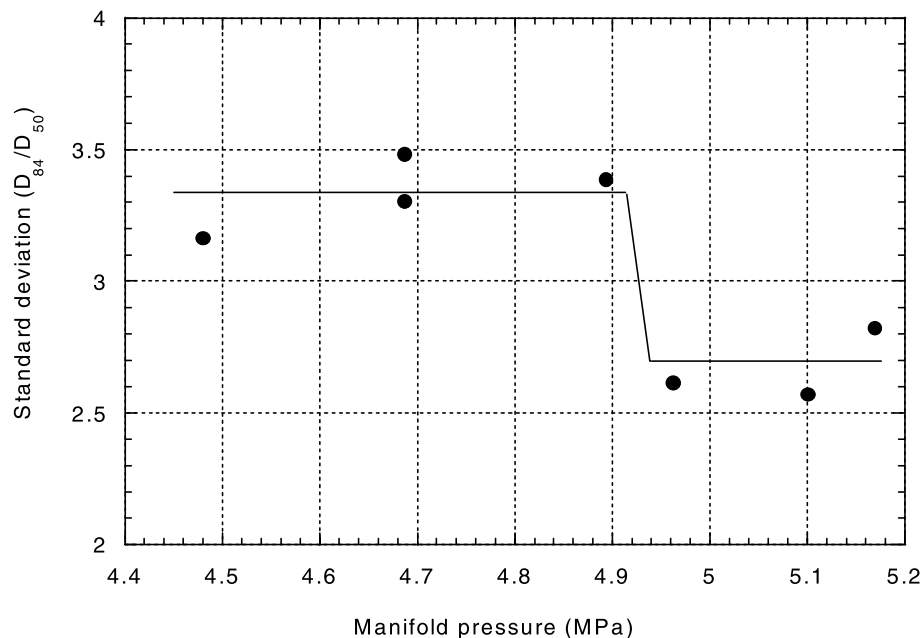


Fig. 5. Variation in standard deviation versus manifold pressure. (Reproduced with modification from [8]).

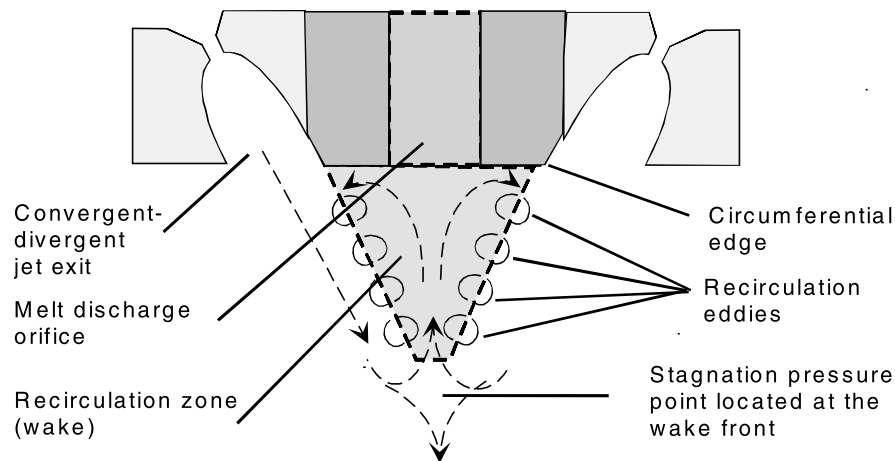


Fig. 6. Schematic of the recirculation zone showing the general gas features of the control model on a central cross-section view of a convergent-divergent discrete jet atomization nozzle. (Reproduced with modification from [11]).

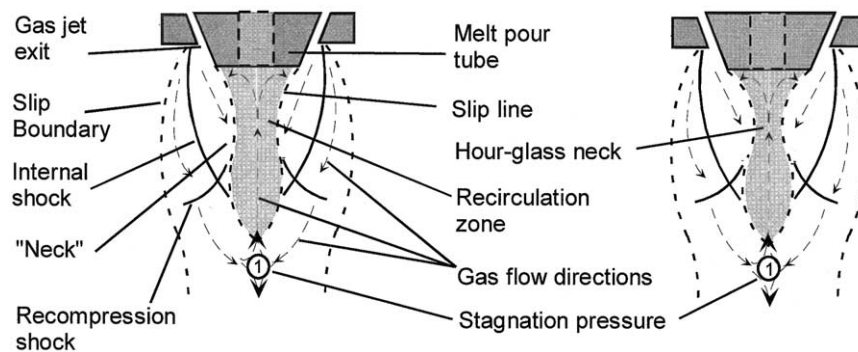


Fig. 7. General gas flow structures in gas-only operation of the HPGA nozzle. Stagnation point 1 forms at the wake front. (Reproduced with modification from [8]).

Fig. 8. The general gas flow features just prior to WCP, showing more constricted necking of the recirculation wake. (Reproduced with modification from [8]).

velocity component at the wake front results in high stagnation pressure. Since the aspiration pressure is a measure of the force of the total recirculating gas flow acting against the melt orifice area, the proposed hypothesis [11] states that the aspiration pressure is the measure of the timed-averaged mass balance of the gas entering and leaving the control volume. Considering that pressure is proportional to the density of gas, the stagnation pressure at the wake front directly contributes to the mass of gas entering the control volume and to the resulting aspiration pressure measured at the melt orifice.

In the gas-only operation below WCP, the external gas flow issuing from an annular convergent (or parallel) jet orifice forms internal and recompression shocks as shown in Fig. 7. This rapidly expanding gas flow pinches the open-wake recirculation zone forming an hour-glass shape with a neck at its mid-section. The stagnation pressure at point 1 feeds into the recirculation zone and affects the aspiration pressure. At a pressure just below WCP, the wake is pinched further, forming a narrower

neck (Fig. 8), but recirculation gas from point 1 can still reach the melt orifice to affect the aspiration pressure.

At pressures above or just above WCP, however, the rapidly expanding gas sections-off and reduces the size of the recirculation zone. This result causes high velocity gas to flow parallel along the centerline away from the recirculation zone (Fig. 9). Since the magnitude of

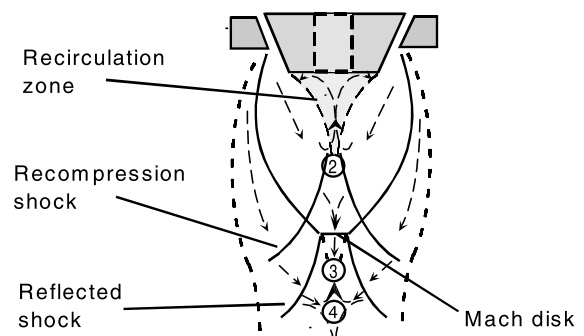


Fig. 9. The general gas flow features just above WCP, showing the formation of Mach disk. Stagnation are labeled points 2, 3, and 4. (Reproduced with modification from [8]).

the radial component of the gas velocity at stagnation point 2 is significantly smaller than that of point 1, the stagnation pressure at point 2 would also be smaller than point 1. In order for the centerline gas pressure to match the pressure down stream of the internal and recompression shocks, the centerline gas has to undergo a shock formation that gives rise to the Mach disk (also known to be a normal shock feature) seen in Fig. 9. This close-wake condition has been observed in both flow visualization imaging [6,11,12] and computational modeling [5,13]. The new recirculation now in closed-wake condition is not only smaller in size but also weaker in strength compared to open-wake conditions, because the stagnation pressure at point 2 is lower than that at point 1 (Fig. 8). Below the Mach disk, the gas immediately decelerates, creating a subsonic high pressure flow downstream, before it reaccelerates further down stream. In addition, the recompression and reflected shock formed at WCP causes the atomizing gas to rotate in towards the centerline of the gas flow field, resulting in the development of two distinct stagnation pressure points downstream of the Mach disk (depicted as point 3 and 4 in Fig. 9). It is therefore conceivable that the magnitude of the stagnation pressure at point 4, operating just above WCP, is significantly larger in magnitude than that at point 1 formed just below WCP. The stagnation pressure at 4 is high (Fig. 9) because the decelerated gas flow downstream of the Mach disk causes a rise in the local pressure at point 4. This pressure rise, furthermore, superimposes more pressure that already exist from the radial component of the gas colliding at point 4. The recirculation zone upstream adjacent the melt orifice, however, is unaffected by the gas pressure at stagnation point 4. This is because the Mach disk acts as a barrier and prevents any backward flow of gas towards the recirculation zone. Thus the diminished recirculation zone, above WCP, is only 'fed' by a weak stagnation pressure at point 2, which in terms causes one to detect a deep aspiration pressure at WCP condition.

The presence of an aspiration hysteresis in this study not only reveals a region of atomization instability, but it also indicates that the creation of the wake closure phenomenon requires the atomizing gas overcoming an energy barrier — in another word, an energy of formation for the wake closure. The argument for the presence of an energy barrier, that the atomizing gas flow has to overcome, is further varified by the fact that the wake closure pressure is very abrupt at long pour tube length. The abruptness of the onset of wake closure and the absence of a wide hysteresis behavior only demonstrates that at high atomizing pressure the atomizing gas energy closely matched the energy of formation. Therefore, a gradual transition to establish wake-closure is absent. On the other hand, at low wake closure pressure that occurs with short pour tube length, once

the take closure is formed during manifold pressure ramp-up, in order to overcome the energy barrier during manifold pressure ramp-down, the gas flow experiences the same energy barrier necessitating the break up the wake closure features. The wake closure can be seen stable at 0.25 MPa below the first wake closure. The hysteresis in the low WCP condition suggests that the wake-closure formation energy, including the establishment of a Mach disk with attached oblique shocks, can be greater than the kinetic energy of the atomizing gas. For example, in order to transition from open-wake to closed-wake at WCP, the atomizing gas has to reverse the flow of the recirculating gas in the wake, and then has to re-accelerate the reversed gas flow to supersonic velocity from point 2 (Fig. 9) to lead to the formation of the Mach disk down stream.

3.2. Atomization model

In gas-only flow conditions during aspiration measurements, the gas flow field surrounding the nozzle accommodates itself via dynamic gas compression and expansion to modulate the pressure between the high velocity gas and the ambient surroundings. Introducing liquid metal into the wake, however, during atomization upsets the gas equilibrium. The presence of incompressible liquid in the wake acts against the compressible gas, disturbing and displacing the established gas dynamic flow features.

Even though most close-coupled atomization practices initiates the flow of liquid metal before the atomizing gas, one will describe the atomization process commencing with a established gas flow field for the sake of clarity. Albeit, the initial melt flow is only a transient state in the whole atomization process.

When initiating liquid atomization at a pressure just below WCP the melt is drawn into the wake by the shallow aspiration in the open wake configuration. This is depicted in Fig. 10a–b showing the melt transition. As more melt flows into the wake, the physical boundary of the melt asserts itself on the external gas flow that opens-up the wake region further (see Fig. 10c). In this new 'opened'-wake configuration, more gas can enter the wake via the enlarged stagnation front, leading to an increase in recirculation gas pressure that could retard the melt flow. As the melt flow slows, the physical boundary asserted by the melt on the gas is momentarily removed, and the wake geometry returns to its original shape that is depicted in Fig. 10b. Then the earlier discussed behaviors repeat themselves. In state 10-c, the recirculation gas may be stronger than that of in state 10-a or 10-b, slowing the melt flow; while in state 10-b the recirculation gas is weak, encouraging aspiration and increasing the melt flow. Due to these changes in gas dynamics, the recirculating gas acts as melt feed control, and the atomization process oscil-

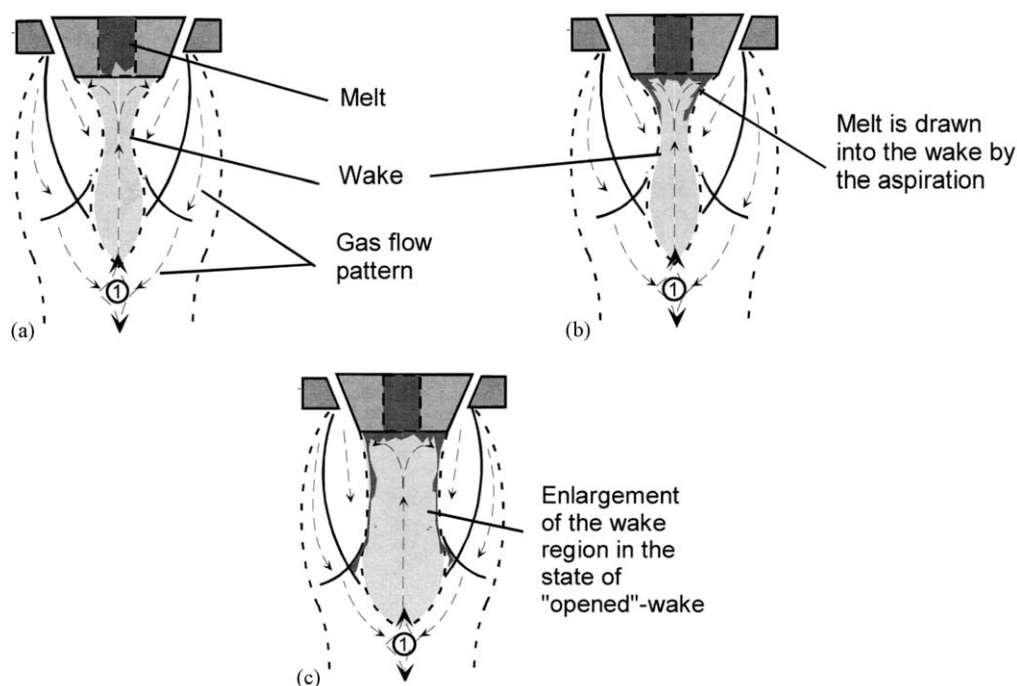


Fig. 10. (a) This shows the gas-only flow features at below WCP just prior to the introduction of the melt into the wake. (b) This shows the melt drawn into the wake by the shallow aspiration. (c) This shows the enlargement of the wake in the state of 'opened'-wake. (Reproduced with modification from [8]).

lates between the two states 10-b and 10-c. Complete wake-closure, however, never occurs at operations below WCP.

When atomizing at or above WCP the melt is readily drawn into the wake by the deep aspiration that develops at wake-closure (Fig. 11a). The atomization transitions from Fig. 11a–b. As the presence to melt increases in the wake, the closed-wake state changes to the 'opened'-wake state and the Mach disk vanishes. This is depicted in Fig. 11c. The physical boundary asserted by the melt upsets the closed-wake gas dynamics, permitting the high pressure gas from the stagnation front at point 4 to rapidly enter the 'opened'-wake (Fig. 11c). The instantaneous surge of very high pressure gas into the wake from the sudden wake opening may temporarily disrupt the melt flow (Fig. 11d and e), leaving the wake momentarily devoid of melt. At this instant, where the physical boundary of the melt is absent, the force from the atomizing gas on the wake neck closes the wake by overcoming the wake-closure formation energy, re-establishing the closed-wake configuration depicted in Fig. 11a. This 'opened' to closed-wake cycle repeats itself until the melt is exhausted. Under any other circumstances where atomization process experiences a large surge in recirculation gas pressure, i.e. overambient aspiration pressure, melt freeze-off may be imminent. In the oscillating closed-wake condition, however, the melt flow into the wake is encouraged by the re-established deep aspiration at the melt orifice. Consequently, the

oscillation between 'opened' and closed-wake scenarios repeat themselves rapidly, leading to what one often observes during melt atomization as a steady pulsation in melt intensity, having fluctuating color and brightness.

4. Discussion

The oscillating behavior hypothesized in the above atomization model indicates that pulsatile atomization mechanism is possibly an inherent feature found in all melt atomization processes. The melt behaves only as a temporary and dynamic solid boundary that adjusts itself continuously during the atomization. From this reasoning, the design for non-oscillating (or quiescent) atomization would be impossible. Several studies have indicated that the flow of liquid metal during atomization may possess a pulsatile characteristic [14–16]. Due to our ability to visually notice the pulsatile behavior during the atomization process suggests that the pulsating frequency has to be much less than 60 Hz. In general, events occurring greater than 60 Hz can not be resolved by the human eye. In high speed film study of rocket fuel atomization, however, Glogowski et al. [14] observed regions of high and low droplet density occurring at wave-like frequency. This observation further substantiates our hypothesized pulsatile atomization mechanism.

The Mach disk formed at WCP, also known as the 'barrel' shock, is an intricate equilibrium gas flow feature. The name 'barrel' shock is coined from its shape resembling a wine barrel. This shape can also be observed in discrete convergent jets gas flow field [17,18]. The equilibrium state can be readily observed in the short pour tube aspiration measurement (see Fig. 1). The fact that the wake-closure can exist at a lower WCP during manifold pressure ramp-down, suggests that wake-closure is a stable equilibrium gas feature. Undoubtedly, the creation of the normal Mach disk, oblique shocks, and reflective shocks require significant energy. Concomitantly, however, is that the entropy of the system increases, resulting from the supersonic gas crossing the Mach disk and oblique shocks that decelerate the supersonic gas before the shock to subsonic velocity after the shocks. This is necessary to accommodate the rapid rise in pressure and temperature across

the shock structures. Associating with the entropy rise is the lowering the equilibrium state of the whole atomizing gas system. Thus a stable wake-closure is formed. The relation between gas velocity and entropy can be expressed in the following equation:

$$e^{-\Delta S/R} = \left[\frac{\{(\gamma+1)/2\} M^2}{1 + \{(\gamma-1)/2\} M^2} \right]^{\gamma/\gamma-1} \left[\frac{2\gamma}{(\gamma+1)} M^2 - \left(\frac{\gamma-1}{\gamma+1} \right) \right]^{1/(1-\gamma)} \quad (1)$$

where ΔS is the change in entropy across the shock at Mach velocity M , R is the universal gas constant, and γ is the gas constant (for argon it is 1.67). It can be seen that entropy change at a negative exponential rate as Mach velocity increase. Therefore, once the wake-closure is established, the entropy of wake-closure will ensure that the wake-closure phenomenon exists at all

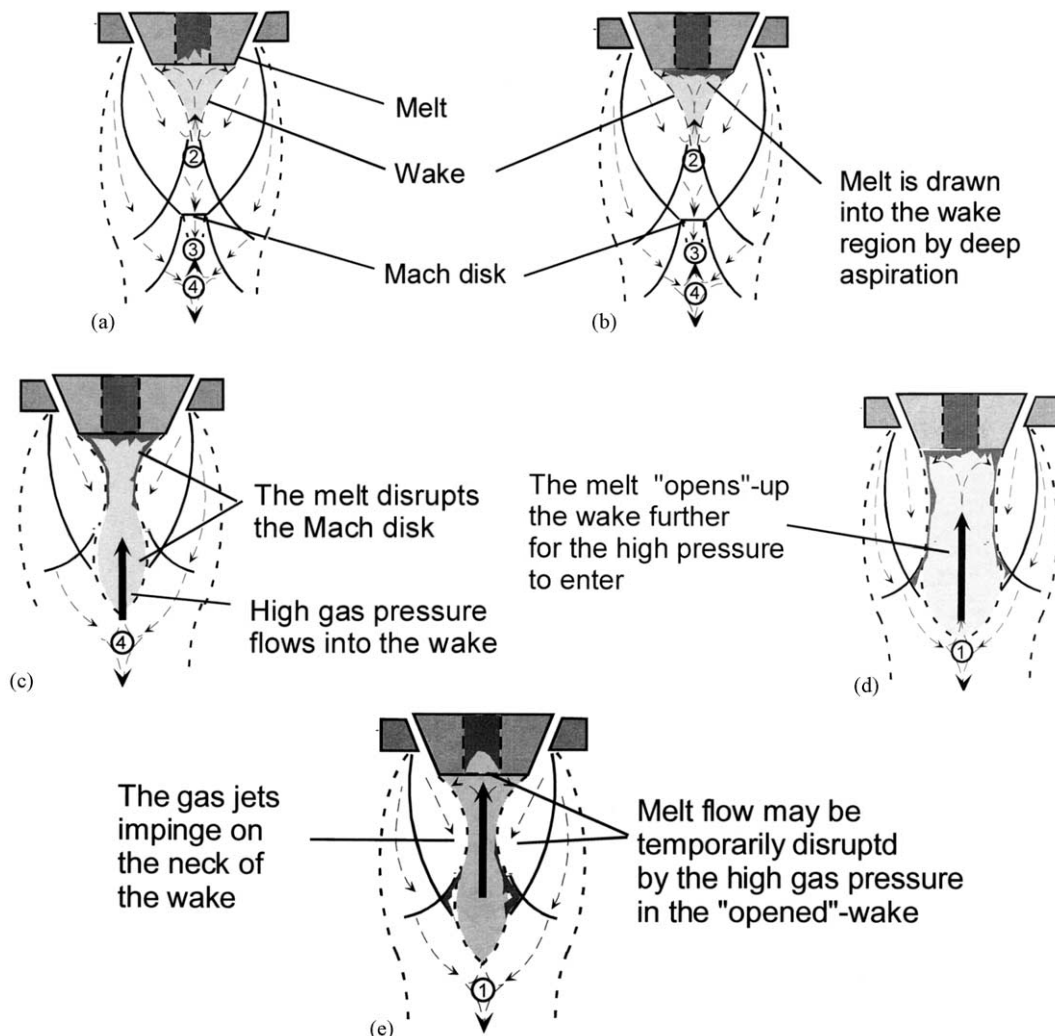


Fig. 11. This shows the gas flow in the closed-wake state just prior to the introduction of melt into the wake. (b) This shows the introduction of the melt drawn in by the deep aspiration at wake-closure. (c) This shows the melt filming the melt tip and the wake. (d) It shows the 'open'-wake condition after the melt has disrupted the Mach disk. The high pressure from stagnation point 4 surges into the wake. (e) It shows the high pressure surge temporarily disrupts the melt flow. The wake begins to neck leading to the reformation of the wake-closure shown in (a). (Reproduced with modification from [8]).

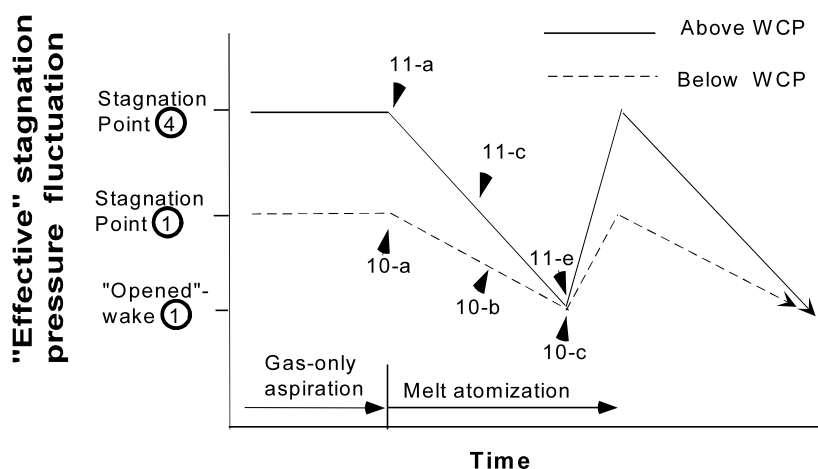


Fig. 12. A schematic depiction of the time averaged fluctuation of the stagnation pressure located at the wake front during gas-only and melt atomization. Some of the conditions given in the figure are from Figs. 10 and 11. (Reproduced with modification from [8])

pressures above the WCP. This explains the unavoidable presence of wake-closure at extremely high operating pressure.

Due to the inherent stability of the wake-closure phenomenon and its associated Mach disk and oblique shocks, this state of equilibrium gives rise to the formation of a hysteresis in aspiration curve (Fig. 1). Therefore, it could be said that the larger the hysteresis the more stable is the wake-closure structure. The Mach disk in the structure is attached along its edges to the internal shock that extends in from the external gas flow. Without the continuity between the internal shock and the normal Mach disk features, the Mach disk structure cannot be at equilibrium and exist. Thus, small perturbations to the wake, changing the presence of the internal and recompression shocks, would drastically result in the disruption of the Mach disk. Experimental evidence by high speed Schlieren photography showed that the normal Mach disk was disrupted by introducing molten tin [12]. Introducing the gas into the recirculation zone has also disrupted the equilibrium state of the Mach disk as well, displacing the surrounding gas flow features and widening the wake region [12,19–22]. Therefore, temporary ‘opened’-wake conditions at above and also below WCP have to exist in gas atomization once the melt is introduced.

The apparent differences in the GMR when atomization above and below WCP are the result of the differences in pressure located at the stagnation points during that momentary ‘opened’-wake conditions. The reduced melt flow rate during atomization above WCP is the result of melt being held back by the strong recirculating gas’s upward force originating from point 4 (Fig. 11d and e), slowing (or may even temporarily halting) the melt stream from the melt orifice. By comparison, the melt flow rate below WCP is relatively unaffected by the stagnation pressure change from

point 1 during melt oscillation. If one was to depict the changes in stagnation pressure at point 1 and 4 for conditions below and above WCP, respectively, in Fig. 12, the timed-averaged stagnation pressure at above WCP would be greater than the timed-averaged pressure at below WCP. Due to this, at above WCP, the melt orifice sees a higher effective timed-average recirculation pressure (or force) than below WCP, which acts against the melt flow, slowing the melt flow rate. Consequently, two distinctive melt flow rates are observed above and below WCP, even though they have practically identical atomization pressures (gas flow rate) and yet very different GMR area observed.

When atomizing above WCP, the observed reduction of the particle standard deviation indicates that the melt filming mechanism may occur. The mechanism may be assisted by the rapid surge of pressure (force) from the stagnation from when the melt disrupted the closed-wake state (Fig. 11d). The upward force of gas into the ‘opened’-wake would act against the downward flowing melt, causing melt to film along the melt tip. In a rocket fuel atomization study, a significant change in the behavior of the spray during high atomizing gas pressure was noticed [13]. It is our belief that the transition from a *nonresonant* mode of operation to a resonant mode at a specific atomization pressure, as described by Glogowski et al. [14], is analogous to the our observed WCP in this study. It was described by Glogowski et al. [14] that at the pressure of resonance, the atomization spray shifted from solid cone to a hollow cone. One is led to believe this corresponds to the observed solid core melt atomization [12] and melt filming atomization [23] conditions, respectively, reported in melt atomization literature. Rocket fuel atomization showed that melt filming occurs only in a resonant mode [14], where a significant portion of the atomized particles were found along the outer edges of

the atomization plume, while the majority of the atomization occurred in the centerline during *nonresonant* mode. This reveals that the melt may enter the wake in two distinct forms during atomization: (1) predominantly as a solid melt core when atomizing below WCP; or (2) principally as a melt film on the melt tip when atomizing above WCP. These two distinct forms are of course highly dependent upon density, flow rate, viscosity and surfaced tension of the melt, and its interaction with upward flowing gas in the wake.

Therefore, two distinct atomization mechanisms may be occurring at any one time during atomization depending on the operating pressure of the atomizing nozzle. Atomizing at pressures below WCP, the atomization is dominated by the secondary melt breakup mechanism, where the solid melt core entering the wake is atomized as if it was experiencing pseudo-free-fall atomization by direct gas impact. The difference in this pseudo-free-fall atomization mechanism compared to the traditional free-fall atomization, is that the gas is brought geometrically closer to the melt core for intensified melt disintegration. Atomizing at pressures above WCP, the atomization is assisted by the primary melt breakup mechanism known to be melt filming. This dominant melt disintegration mechanism above WCP is supplemented by the secondary breakup mechanism that occurs outside the wake down stream. This melt filming breakup mechanism is most likely to have occurred as the sheet-to-ligament break-up disintegration process suggested by Dombrowski and Johns [24]. As it was demonstrated in this study, Heat L1271, which was atomized above WCP, generated a narrower particle size distribution than Heat L1270, which was atomized below WCP, even though both Heats were atomized at pressures that were within 1.4% of one another.

5. Conclusions

Using molten metal atomization, this study has demonstrated that deep aspiration (above wake-closure pressure) does not translate into an increase in molten metal flow rate. On the contrary, WCP condition slows down the melt flow. Therefore, atomization rates can not be related to the depth of the gas-only aspiration pressures. This study confirmed the work by Miller et al. [7] who used water atomization to show that there is little correlation between aspiration pressure and liquid flow rate. Atomizing at above WCP indeed produce the finest powder but this is achieved at the expense of higher GMR compared with below WCP atomization.

This study has demonstrated experimentally that atomizing at pressures just 0.014 MPa (2 psi) above the WCP increases the yield of finer particles, compared to yields obtained at 0.034 MPa (5 psi) below WCP by 42% or by 26.5 microns.

The pressure at which wake-closure occurs around 4.94 MPa (715 psi) in the industrial atomizer nozzle is highly repeatable. The occurrence of wake-closure, of course, is highly specific to the nozzle designs and setups. In the short pour tube configuration, two wake closures were observed at 4.35 and 4.1 MPa, during manifold pressure ramp-up and ramp-down, respectively. Thus suggesting that the wake closure phenomenon is an equilibrium gas flow feature that has an energy of formation.

A gas dynamic model is presented describing the formation of wake closure during deep aspiration. The model showed that a reduced recirculation wake size corresponds to a weaker recirculation strength, which would give rise to the deep aspiration pressure measurement that one has observed in the study.

A pulsating atomization model is presented using gas dynamics to explain the mechanics of the atomization processes occurring above and below WCP. The model shows that stable wake-closure cannot be sustained during melt atomization. And that the stagnation front pressure at wake-closure facilitates melt filming during the 'opened'-wake state of the pulsatile atomization process. This melt filming process at above WCP gives rises to finer particle size distribution, lower melt flow rate, and higher GMR. On the other hand, operating below WCP, the melt breakup mechanism could be similar to that of the free-fall atomization process, where the melt is introduced into the wake as a solid core and it is atomized mostly by impact collision. This process at below WCP is less efficient than melt filming at above WCP and therefore gave rise to larger mean particle sizes and wider standard deviations.

This study has presented a model that has elucidated the causes of why one atomizer condition yielded finer particle size distribution when operating above WCP compared to yields obtained when atomizing below WCP, even though the two atomization pressures may be different by less than 1.5% of each other.

Finally, this paper has presented a hypothesis for a pulsatile atomization model that may have elucidated the association between the measured gas-only aspiration pressures and its effects and influences on the powder yields obtained in the melt atomizations operated around WCP.

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References

- [1] A. Yule, J.J. Dunkley, *Atomization of Metals*, Oxford University Press, Oxford, UK, 1994.
- [2] I.E. Anderson, R.S. Figliola, in: P.U. Gummesson, D.A. Gustafson (Eds.), *Modern Developments in Powder Metallurgy*, vol. 19, MPIF, Princeton, NJ, 1988, p. 205.
- [3] R.R. Raman, U.S. Patent No. 4,778,516.
- [4] I.E. Anderson, U.S. Patent No. 5,125,574.
- [5] P.I. Espina, S.D. Ridder, F.S. Biancaniello, G.E. Mattingly, Aerodynamic analysis of the aspiration phenomena, in: E.J. Lavernia, H. Henein, I.E. Anderson (Eds.), *Characterization and Diagnostics of Ceramic and Metal Particulate Processing*, TMS, Warrendale, PA, 1989, p. 49.
- [6] I.E. Anderson, R.S. Figliola, H. Morton, *Mat. Sci. Eng. A148* (1991) 101.
- [7] R.S. Miller, et al., *Int. J. Powder Metall.* 32 (4) (1996) 341.
- [8] J. Ting, M.W. Peretti, W.B. Eisen, Control of fine powder production and melt flow rate using gas dynamics, in: H. Ferguson, D.T. Whychell (Eds.), *Advances in Powder Metallurgy & Particulate Materials – 2000*, APMI-MPIF, Princeton, NJ, 2000, pp. 2.27–2.40.
- [9] G. Wolf, A. Lang, H.W. Bergmann, Investigation on melt atomization with gas and liquefied cryogenic liquid, in: K. Bauckhage, V. Uhlenwinkel, U. Fritsching (Eds.), *Spray deposition and Melt Atomization*, vol. 2, 2000, pp. 535–547.
- [10] W.P. Sule, T.J. Mueller, *J. Spacecraft* 10 (11) (1983) 521–525.
- [11] J. Ting, I.E. Anderson, R. Terpstra, J. Mi, Design and testing of an improved convergent-divergent discrete-jet high pressure gas atomization nozzle, in: J.J. Oakes, J.H. Reinshagen (Eds.), *Advances in Powder Metallurgy & Particulate Materials – 1998*, Part 10, vol. 3, APMI-MPIF, Princeton, NJ, 1998, pp. 25–39.
- [12] S. Mates, *Gas Atomization of Molten Metal*, PhD Thesis, Pennsylvania State University, 1999.
- [13] P.I. Espina, U. Piomelli, Numerical Simulation of the Gas Flow in Gas-Melt Atomizers, *Proceedings of 1998 ASME Fluid Engineering Division Summer Meeting*, 21–25 June, 1998, Washington, DC. (FEDSM98-4901) p. 1.
- [14] M. Glogowski, M. Bar-Gill, C. Puissant, T. Kaltz, M. Milicic, M. Micci, *Shear coaxial Injector Instability Mechanisms*, 30th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, 27–29 June, 1994, Indianapolis, IN.
- [15] S.D. Ridder, S.A. Osella, P.I. Espina, F.S. Biancaniello, *Int. J. Powder Metall.* 28 (2) (1992) 133–147.
- [16] Personal observations at Crucible Research.
- [17] J. Ting, *Gas Atomization Processing of Tin and Silicon Modified LaNi₅ for Nickel-Metal Hydride Battery Applications*, PhD, Iowa State University, Ames, IA, 1998.
- [18] U. Heck, U. Fritsching, K. Bauckhage, *Atomization Sprays* 10 (2000) 25–46.
- [19] J.L. Herrin, J.C. Dutton, *J. Spacecraft Rockets* 31 (6) (1994) 1021.
- [20] T. Mathur, J.C. Dutton, *J. Spacecraft Rockets* 33 (1) (1996) 30.
- [21] K. Issac, et al., *AIAA J.* 32 (8) (1994) 1640.
- [22] T. Mathur, J.C. Dutton, *AIAA J.* 34 (6) (1996) 1153.
- [23] A. Unal, *Metall. Trans. B* 20 (2) (1989) 61–69.
- [24] N. Dombrowski, W. Johns, *Chem. Eng. Sci.* 18 (1963) 203–214.